(FP1) 1. Roots of polynomial equations HW

NO. 1

The cubic equation $x^3 + bx^2 + cx + d = 0$, where b, c and d are constants, has roots α, β, γ. It is given that $\alpha\beta\gamma = -1$.

a. State the value of d. [1]

b. Find a cubic equation, with coefficients in terms of b and c, whose roots are $\alpha+1$, $\beta+1$, $\gamma+1$. [3]

c. Given also that $\gamma + 1 = -\alpha - 1$, deduce that $(c - 2b + 3)(b - 3) = b - c$. [4]

A level FURTHER MATHEMATICS 2020/Winter/Paper1_2/Q1

Mark scheme

Question	Answer	Marks	Guidance
a	$d = 1$	B1	
		1	
b	$y = x+1 \Rightarrow x = y-1$	**B1**	Uses correct substitution.
	$y^3 + (b-3)y^2 + (c-2b+3)y + b - c = 0$	**M1** **A1**	Substitutes and expands.
	Alternative method for question 1(b)		
	$(\alpha+1)(\beta+1) + (\alpha+1)(\gamma+1) + (\beta+1)(\gamma+1) = c-2b+3$	**B1**	
	$(\alpha+1+\beta+1+\gamma+1) = 3-b$, $(\alpha+1)(\beta+1)(\gamma+1) = c-b$	**M1**	Using these relationships.
	$y^3+(b-3)y^2 + (c-2b+3)y + b - c = 0$	**A1**	
		3	
c	$\beta+1 = -(b-3)$	**B1**	Uses sum of roots.
	$-(\alpha+1)(\alpha+1) = c-2b+3$	**B1**	Uses sum of products in pairs.
	$-(\alpha+1)(\beta+1)(\alpha+1) = -(b - c)$	**M1**	Applies product of roots.
	$\Rightarrow (c-2b+3)(b-3) = b - c$	**A1**	AG
		4	

Explanation

a.

Compare coefficients of $(y - a)(y - \beta)(y - \gamma) = 0$ and $x^3 + bx^2 + cx + d = 0$

$d = -\partial\beta\gamma = 1$ **B1**

b)

use substitution y=x+1, then x=y-1 **B1**

$$(y-1)^3 + b(y-1)^2 + c(y-1) + 1 = 0$$

Expand binomial terms

$$y^3 - 3y^2 + 3y - 1 + by^2 - 2by + b + cy - c + 1 = 0$$

$$y^3 + (b-3)y^2 + (3-2b+c)y + (b-c) = 0$$ **M1 A1**

c)

From equation in part b), we have

$$(a+1) + (\beta+1) + (\gamma+1) = -(b-3)$$

And$\gamma + 1 = -a - 1$

So$\beta + 1 = -(b-3) = -b + 3$ **B1**

$\beta + 1$ is a root of equation in part b), use y=$\beta + 1$= -b+3 **B1**

$$(-b+3)^3 + (b-3)(-b+3)^2 + (3-2b+c)(-b+3) + (b-c) = 0$$ **M1**

First two terms cancelled, then we have required equation:

$$(c-2b+3)(b-3) = b - c$$ **A1**

Comment

a. Very few candidates made a sign error in writing down d.

b. The most efficient solutions came from using the substitution $x = y - 1$. Some candidates returned to the original equation and used its root relationships to find the required coefficients for the new equation. This could become very complicated and risked introducing errors. In either case, the most common error was to forget to use $d = 1$ from **(a)**

c. Candidates were asked to prove a given result and had to ensure that all necessary steps were clearly shown. Those candidates who wrote down the three relationships between the coefficients in **(b)** and the roots were far more likely to have correct signs and to make the structure clear. Then using the given relationship $\gamma+1 = -\alpha - 1$ would simplify the expressions and make the required result quickly obtainable. Those candidates who tried to multiply brackets, or to substitute directly, found it difficult not to lose terms or signs.

NO. 2

The cubic equation $7x^3 + 3x^2 + 5x + 1 = 0$, has roots α, β, γ.

a. Find a cubic equation whose roots are α^{-1}, β^{-1}, γ^{-1}. [3]

b. Find the value of $\alpha^{-2} + \beta^{-2} + \gamma^{-2}$. [2]

c. Find the value of $\alpha^{-3} + \beta^{-3} + \gamma^{-3}$. [2]

A level FURTHER MATHEMATICS 2020/Summer/Paper1_3/Q1

Mark scheme

Question	Answer	Marks
a	$y = x^{-1}$	B1
	$7y^{-3} + 3y^{-2} + 5y^{-1} + 1 = 0 \Rightarrow y^3 + 5y^2 + 3y + 7 = 0$	M1 A1
		3
b	$\alpha^{-2} + \beta^{-2} + \gamma^{-2} = (-5)^2 - 2(3) = 19$	M1 A1
c	$\alpha^{-3} + \beta^{-3} + \gamma^{-3} = -5(19) - 3(-5) - 21 = -101$	M1 A1
		4

Explanation

a.

Use substitution: $y = x^{-1}$, then $x = \frac{1}{y}$ **B1**

$$7\frac{1}{y^3} + 3\frac{1}{y^2} + 5\frac{1}{y} + 1 = 0$$

Multiply by y^3

$y^3 + 5y^2 + 3y + 7 = 0$ **M1 A1**

b)

Compare coefficients of $y^3 + 5y^2 + 3y + 7 = 0$ and $(y - a^{-1})(y - \beta^{-1})(y - \gamma^{-1}) = 0$

$$a^{-1} + \beta^{-1} + \gamma^{-1} = -5$$

$$\alpha^{-1}\beta^{-1} + \alpha^{-1}\gamma^{-1} + \beta^{-1}\gamma^{-1} = 3$$

Use formula :$a^2 + b^2 = (a + b)^2 - 2ab$

$a^{-2} + \beta^{-2} + \gamma^{-2} = (\Sigma\alpha^{-1})^2 - 2\Sigma a^{-1}\beta^{-1} = (-5)^2 - 2 \times 3 = 19$ **M1 A1**

c)

$$y^3 + 5y^2 + 3y + 7 = 0$$

Replace y with $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ and obtain equations like:

$$a^{-3} + 5\alpha^{-2} + 3\alpha^{-1} + 7 = 0$$

Sum three equations together:

$$\Sigma a^{-3} + 5\Sigma\alpha^{-2} + 3\Sigma\alpha^{-1} + \sum 7 = 0$$

$\Sigma\alpha^{-2}$ and $\Sigma\alpha^{-1}$ are obtained in b)

$\alpha^{-3} + \beta^{-3} + \gamma^{-3} = \Sigma a^{-3} = -5 \times 19 - 3 \times (-5) - 3 \times 7 = -101$ **M1 A1**

NO. 3

The roots of the cubic equation $x^3 + 2x^2 - 3 = 0$ are α, β and γ.

i. By using the substitution $y = \frac{1}{x^2}$, find the cubic equation with roots $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$. [3]

ii. Hence find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. [1]

iii. Find also the value of $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\gamma^2\alpha^2}$. [1]

A level FURTHER MATHEMATICS 2017/Summer/Paper1_3/Q1

Mark scheme

Question	Answer	Marks	Guidance
1.i	$y=\frac{1}{x^2}\Rightarrow x=\frac{1}{\sqrt{y}}$	**M1**	Rearranging to make x the subject
	$\frac{1}{y\sqrt{y}}+\frac{2}{y}-3=0\Rightarrow\frac{1}{y\sqrt{y}}=3-\frac{2}{y}$	**M1**	Substituting and squaring
	$\Rightarrow 9y^3-12y^2+4y-1=0$ SR **B1** for finding cubic by manipulating roots	**A1**	OE
	Total:	**3**	
ii	$\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}=\frac{12}{9}$ or $\frac{4}{3}$	**B1FT**	
	Total:	**1**	
iii	$\frac{1}{\alpha^2\beta^2}+\frac{1}{\beta^2\gamma^2}+\frac{1}{\gamma^2\alpha^2}=\frac{4}{9}$	**B1FT**	
	Total:	**1**	

Explanation

1.i

$y=\frac{1}{x^2}$, $x=\frac{1}{\sqrt{y}}$ **M1**

(± is not necessary since we will square later)

$\left(\frac{1}{\sqrt{y}}\right)^3+2\left(\frac{1}{\sqrt{y}}\right)^2-3=0$

Multiply by $y^{\frac{3}{2}}$

$1 + 2\sqrt{y} - 3y\sqrt{y} = 0$ **M1**

$1 = \sqrt{y}(3y - 2)$

$1 = y(3y - 2)^2$

$1 = y(9y^2 - 12y + 4)$

The cubic equation is

$9y^3 - 12y^2 + 4y - 1 = 0$ **A1**

ii.

compare the coefficients of

$9y^3 - 12y^2 + 4y - 1 = 0$

and

$9\left(y - \frac{1}{\alpha^2}\right)\left(y - \frac{1}{\beta^2}\right)\left(y - \frac{1}{\gamma^2}\right) = 0$

then:

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{12}{9} = \frac{4}{3}$ **B1FT**

iii.

$$\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\gamma^2\alpha^2} = \frac{4}{9}$$ **B1FT**

Comment

The majority of candidates used the correct method and were able to substitute for x as requested. Good candidates rearranged the equation correctly before squaring so that they could eliminate the fractional powers and reach a cubic equation. Successful candidates followed the instruction (hence) and used their cubic to answer **(ii)**. Some opted to use the properties of roots to answer the final part.

Answers: **(i)** $9y^3 - 12y^2 + 4y - 1 = 0$ **(ii)** $\frac{12}{9}$ or $\frac{4}{3}$ **(iii)** $\frac{4}{9}$

NO. 4

By finding a cubic equation whose roots are α, β and γ, solve the set of simultaneous equations

$$\alpha + \beta + \gamma = -1,$$
$$\alpha^2 + \beta^2 + \gamma^2 = 29,$$
$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -1.$$

[8]

A level FURTHER MATHEMATICS 2017/Summer/Paper1_1/Q7

Mark scheme

Question	Answer	Marks	Guidance
7	$2\Sigma\alpha\beta = 1 - 29 \Rightarrow \Sigma\alpha\beta = -14$	M1A1	
	$\frac{\sum\alpha\beta}{\alpha\beta\gamma} = \frac{-14}{\alpha\beta\gamma} = -1 \Rightarrow \alpha\beta\gamma =$ -14	M1A1 FT	
	$\Rightarrow x^3 + x^2 - 14x - 14$	A1	
	$\Rightarrow (x + 1)(x^2 - 14)$	M1A1	Attempt to factorise cubic
	$\Rightarrow$Solution is –1, in $\pm\sqrt{14}$ any order. Accept ±3.74 (awrt) SR **B1** for correct roots without working	A1	
	Total:	8	

Explanation

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$29 = (-1)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$\alpha\beta + \alpha\gamma + \beta\gamma = -14$ **M1A1**

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + a\beta}{a\beta\gamma} = -1$

$\frac{-14}{a\beta\gamma} = -1$

$a\beta\gamma = 14$ **M1A1**

The cubic equation is:

$(x - a)(x - \beta)(x - \gamma) = 0$

$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - (a\beta\gamma) = 0$

$x^3 + x^2 - 14x - 14 = 0$ **A1**

(Try factors (x±1),(x±2),(x±7),(x±14))

$(x + 1)(x^2 - 14) = 0$ **M1A1**

Thus α, β, γ are -1, $\sqrt{14}, -\sqrt{14}$ **A1**

Comment

Candidates were usually able to find the cubic equation from the given equations, though there were a few sign errors. A number of candidates, including some strong ones, stopped after this point and seemed to have forgotten that they were required to find the

roots. Those who did go on were mostly successful in factorising the cubic as $(x + 1)(x^2 - 14)$ and therefore identifying the three roots.

Answer: $x^3 + x^2 - 14x - 14 = 0$, $x = -1, \pm\sqrt{14}$

NO. 5

i. The cubic equation

$$z^3 - z^2 - z - 5 = 0$$

has roots α, β and γ. Show that the value of $\alpha^3 + \beta^3 + \gamma^3$ is 19. [4]

ii. Find the value of $\alpha^4 + \beta^4 + \gamma^4$. [2]

iii. Show that the cubic equation with roots $\frac{\alpha - 1}{\alpha}$, $\frac{\beta - 1}{\beta}$ and $\frac{\gamma - 1}{\gamma}$ may be found using the substitution $z = \frac{1}{1 - x}$, and find this equation, giving your answer in the form $px^3 + qx^2 + rx + s = 0$, where p, q, r and s are constants to be determined. [4]

A level FURTHER MATHEMATICS 2016/Summer/Paper1_3/Q8

Mark scheme

Qu	Solution	Part Marks
8.i	Let S_n denote $\alpha^n + \beta^n + \gamma n$ $S_2 = S_1^2 - 2\Sigma\alpha\beta = 1^2 - 2 \times (-1) = 3$	M1A1
	$S_3 = S_2 + S_1 + 15 = 3 + 1 + 15 = 19$ (**AG**)	M1A1
		[4]
ii	Alt method: $\Sigma\alpha = 1$, $\Sigma\alpha\beta = -1$, and $\alpha\beta\gamma = 5$ **M1, A1** $S_3 = S_1^3 - 3\Sigma\alpha\,\Sigma\alpha\beta + 3\alpha\beta\gamma = 19$ **M1, A1** $S_4 = S_3 + S_2 + 5S_1 = 19 + 3 + 5 = 27$	M1A1
		[2]
iii	$x = \frac{z-1}{z} = 1 - \frac{1}{z} \Rightarrow \frac{1}{z} = 1 - x \Rightarrow z = \frac{1}{1-x}$	B1
	Hence $\frac{1}{(1-x)^3} - \frac{1}{(1-x)^2} - \frac{1}{(1-x)} - 5 = 0$ is the required equation.	M1
	$\Rightarrow 1 - (1 - x) - (1 - x)^2 - 5(1 - x)^3 = 0 \Rightarrow \ldots \Rightarrow 5x^3 - 16x + 18x - 6 = 0$	M1A1
		[4]

Explanation

8.i

Let S_n donotes $\alpha^n + \beta^n + \gamma^n$

From $z^3 - z^2 - z - 5 = 0$

$a\beta\gamma = 5,\ \alpha\beta + \alpha r + \beta r = -1,\ \alpha + \beta + \gamma = 1$

$a^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\ \alpha\beta + \alpha r + \beta r) = 1^2 - 2(-1) = 3$ **M1A1**

α, β and γ are roots of $z^3 - z^2 - z - 5 = 0$

So $a^3 - a^2 - a - 5 = 0$

$S_3 - S_2 - S_1 - 5 \times 3 = 0$

$S_3 = S_2 + S_1 + 5 \times 3 = 3 + 1 + 15 = 19$

M1A1

ii.

$z^3 - z^2 - z - 5 = 0 \rightarrow z^4 - z^3 - z^2 - 5z = 0$

$S_4 - S_3 - S_2 - 5S_1 = 0$

$S_4 = S_3 + S_2 + 5S_1 = 19 + 3 + 5 \times 1 = 27$ **M1A1**

iii.

$z = \frac{1}{1-x} \rightarrow 1 - x = \frac{1}{z} \rightarrow x = 1 - \frac{1}{z} = \frac{z-1}{z}$ **B1**

If α, β and γ are roots of the equation in terms of z

$\frac{\alpha-1}{\alpha}, \frac{\beta-1}{\beta}, \frac{\gamma-1}{\gamma}$ will be the roots in terms of x

$\left(\frac{1}{1-x}\right)^3 - \left(\frac{1}{1-x}\right)^2 - \frac{1}{1-x} - 5 = 0$ **M1**

Multiply by $(1-x)^3$

$$1-(1-x)-(1-x)^2-5(1-x)^3=0$$

$$1-1+x-(1-2x+x^2)-5(1-3x+3x^2-x^3)=0$$

$$5x^3-16x^2+18x-6=0$$ **M1A1**

Comment

This question was mainly well done, although not all candidates who attempted to use a formula, for example for $\Sigma\alpha^3$, remembered it correctly in part **(i)**. They might have had more success tackling the summation using the equation itself, although some managed to manipulate expressions to a successful conclusion. Part **(ii)** was well done. Candidates were required to justify the given substitution in **(iii)**, which many omitted, and this was one of the few questions were algebraic errors were relatively common. Some candidates attempted to transform the given equation in all three parts, but this was rarely time-effective or indeed successful.

Answers: **(ii)** $S_4 = 27$ **(iii)** $5x^3 - 16x + 18x - 6 = 0$

NO. 6

The cubic equation $x^3 + bx^2 + cx + d = 0$, where $d \neq 0$, has roots α, β, γ such that $\gamma = \frac{1}{\beta}$.

a. Show that $\beta + \frac{1}{\beta} = d - b$. [3]

b. Show also that $\beta + \frac{1}{\beta} = \frac{1-c}{d}$. [2]

c. It is given that $b = 3$, $c = -3$ and $d > 0$.

i. Find the value of d. [2]

ii. Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

Mark scheme

Question	Answer	Marks	Guidance
2(a)	$\alpha = -d$	**B1**	SOI
	$\alpha + \beta + \frac{1}{\beta} = -b$	**M1**	Uses $\alpha + \beta + \gamma = -b$ and replaces γ with $\frac{1}{\beta}$. OE
	$\beta + \frac{1}{\beta} = d - b$	**A1**	AG.
		3	

Question	Answer	Marks	Guidance
2(b)	$-d\beta+1-\frac{d}{\beta}=c$	**M1**	Uses $\alpha\beta+\beta\gamma+\gamma\alpha=c$.
	$\beta+\frac{1}{\beta}=\frac{1-c}{d}$	**A1**	AG.
		2	
2(c)(i)	$d-3=\frac{1+3}{d}\Rightarrow d^2-3d-4=0$	**M1**	Equates expressions given in **(a)** and **(b)**, with correct use of coefficients.
	$d=4$	**A1**	$d=4$ only for final answer.
		2	
2(c)(ii)	$\alpha^2+\beta^2+\gamma^2=b^2-2c$	**M1**	$\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)$ with correct use of coefficients.
	15	**A1**	
		2	

NO. 7

The cubic equation $x^3 + 2x + 1 = 0$ has roots α, β, γ.

a. Find a cubic equation whose roots are $\alpha^3 - 1, \beta^3 - 1, \gamma^3 - 1$. [3]

b. Find the value of $(\alpha^3 - 1)^2, (\beta^3 - 1)^2, (\gamma^3 - 1)^2$. [2]

c. Find the value of $(\alpha^3 - 1)^3, (\beta^3 - 1)^3, (\gamma^3 - 1)^3$. [2]

Mark scheme

Question	Answer	Marks	Guidance
a	$y = x^3 - 1$	B1	Uses correct substitution.
	$(y+1)^{\frac{3}{3}} + 2(y+1)^{\frac{1}{3}} + 1 = 0 \Rightarrow 2(y+1)^{\frac{1}{3}} = -y-2$ $8(y+1) = -(y+2)^3 \Rightarrow 8y+8 = -y^3 - 6y^2 - 12y - 8$	M1	Substitutes and obtains an equation not involving radicals.
	$y^3 + 6y^2 + 20y + 16 = 0$	A1	
		3	
b	$(\alpha^3 - 1)^2 + (\beta^3 - 1)^2 + (\gamma^3 - 1)^2 = 6^2 - 2(20)$	M1	Uses $\alpha^3+\beta^3+\gamma^3 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha)$.
	−4	A1	FT
		2	
c	$(\alpha^3 - 1)^3 + (\beta^3 - 1)^3 + (\gamma^3 - 1)^3 = -6(-4) - 20(-6) - 16(3)$	M1	Uses their equation from (a) or a valid formula for the sum of cubes.
	96	A1	
		2	

Comment

a. This was mostly done well. There were two main methods used. Substitution of a surd followed by simplification, or rearrangement to a form including x^3 followed by substitution, were equally successful. Most demonstrated understanding of the need to isolate the surd term prior to cubing.

A few tried using the roots of the original equations to create the roots of the new equation but were largely unsuccessful.

b. The result usually followed correctly from their answer for **part (a)**.

c. Most candidates used $S_3 + 6S_2 + 20S_1 + 16S_0 = 0$ accurately.

A few tried to use formulae but these formulae were often wrongly remembered, or incorrect values were used.

NO. 8

It is given that

$$\alpha + \beta + \gamma + \delta = 2,$$
$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 3,$$
$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 = 4,$$

a. Find the value of $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$. [2]

b. Find the value of $\alpha^2\beta + \alpha^2\gamma + \alpha^2\delta + \beta^2\alpha + \beta^2\gamma + \beta^2\delta + \gamma^2\alpha + \gamma^2\beta + \gamma^2\delta + \delta^2\alpha + \delta^2\beta + \delta^2\gamma$. [3]

c. It is given that α, β, γ, δ are the roots of the equation

$$6x^4 - 12x^3 + 3x^2 + 2x + 6 = 0.$$

i. Find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$. [3]

ii. Find the value of $\alpha^5 + \beta^5 + \gamma^5 + \delta^5$. [2]

Mark scheme

Question	Answer	Marks	Guidance
a	$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$ $3 = 2^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$	M1	Substitutes into formula for sum of squares.
	$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{1}{2}$	A1	
		2	
b	$\alpha^2(\beta + \gamma + \delta) + \beta^2(\alpha + \gamma + \delta) + \gamma^2(\alpha + \beta + \delta) + \delta^2(\alpha + \beta + \gamma)$ $= \alpha^2(2 - \alpha) + \beta^2(2 - \beta) + \gamma^2(2 - \gamma) + \delta^2(2 - \delta)$	M1 A1	Factorises and substitutes.
	$2(3) - 4 = 2$	A1	
		3	
c.i	$6S_4 - 12S_3 + 3S_2 + 2S_1 + 24 = 0 \Rightarrow 6S_4 - 12(4) + 3(3) + 2(2) + 24 = 0$	M1	Sums and substitutes.
		A1	
	$S_4 = \frac{11}{6}$	A1	
		3	
c.ii	$6S_5 - 12S_4 + 3S_3 + 2S_2 + 6S_1 = 0 \Rightarrow$ $6S_5 - 12\left(\frac{11}{6}\right) + 3(4) + 2(3) + 12 = 0$	M1	Multiplies equation through by x, sums and substitutes. $S_5 - 2S_4 + \frac{1}{2}S_3 + \frac{1}{3}S_2 + S_1 = 0$ $\Rightarrow S_5 - 2\left(\frac{11}{6}\right) + \frac{1}{2}(4) + \frac{1}{3}(3) + 2 = 0$
	$S_5 = -\frac{4}{3}$	A1	
		2	

Comment

a. Almost all candidates recalled and applied the correct formula.

b. The simplest solution was to recognise that $\beta + \gamma + \delta = 2 - \alpha$ and so the first three terms can be simplified to $2\alpha^2 - \alpha^3$, for example. Other successful solutions used known formulas. Those candidates who tried to evaluate $\sum\alpha\sum\alpha^2$ often made algebraic mistakes.

c. i. The majority found this straightforward. The summation of the equations was done extremely well, the only common problem was to forget that there were 4 roots.

ii. A correct answer in **(i)** was usually followed by a correct solution here.

NO. 9

The cubic equation $27x^3 + 18x^2 + 6x - 1 = 0$ has roots α, β, γ.

a. Show that a cubic equation with roots $3\alpha+1, 3\beta+1, 3\gamma+1$ is $y^3 - y^2 + y - 2 = 0$. [3]

The sum $(3\alpha+1)^n + (3\beta+1)^n + (3\gamma+1)^n$ is denoted by S_n.

b. Find the values of S_2 and S_3. [4]

c. Find the values of S_{-1} and S_{-2}. [3]

A level FURTHER MATHEMATICS 2023/Winter/Paper1_2/Q4

Mark scheme

Question	Answer	Marks	Guidance
a	$y = 3x + 1 \Rightarrow x = \frac{1}{3}(y-1)$ $\Rightarrow 27\left(\frac{y-1}{3}\right)^3 + 18\left(\frac{y-1}{3}\right)^2 + 6\left(\frac{y-1}{3}\right) - 1 = 0$	B1	Substitutes.
	$\Rightarrow (y-1)^3 + 2(y-1)^2 + 2(y-1) - 1 = 0$ $\Rightarrow y^3 - 3y^2 + 3y - 1 + 2y^2 - 4y + 2 + 2y - 2 - 1 = 0$	M1	Expands.
	$y^3 - y^2 + y - 2 = 0$	A1	AG.
		3	
b	$S_2 = 1^2 - 2(1) = -1$	M1 A1	Uses formula for sum of squares, AG.
	$S_3 = (3\alpha+1)^2 + (3\beta+1)^2 + (3\gamma+1)^2 = -1-(1)+6$	M1	Uses $y^3 = y^2 - y + 2$ or expands and uses original equation.
	4	A1	
		4	
c	$S_{-1} = \frac{(3\alpha+1)(3\beta+1)+(3\beta+1)(3\gamma+1)+(3\gamma+1)(3\alpha+1)}{(3\alpha+1)(3\beta+1)(3\gamma+1)} = \frac{1}{2}$	B1	
	$2S_{-2} = S_1 - 3 + S_{-1} = 1 - 3 + \frac{1}{2}$	M1	Uses $2y^{-2} = y - 1 + y^{-1}$.
	$S_{-2} = -\frac{3}{4}$	A1	CAO
		3	

Comment

A few candidates became confused between equations but the methods are, in general, well-understood.

a. Most used the substitution method and showed the expansion of $(y-1)^3$ fully to justify their arrival at the given answer. Some found the coefficients of the new equation using the sums and products from the original equation and this also worked effectively in this case.

b. As long as they used the right equation the sum of the squares of the roots was found accurately using the standard method.

To find S_3 most used the relationship $S_3 - S_2 + S_1 - 2S_0 = 0$ and errors were usually caused by saying $S_0 = 1$.
A few tried to use formulae but often remembered them incorrectly.

c. The value of S_{-1} was mostly correct, either from the coefficients of the equation or from the relationship $S_2 - S_1 + S_0 - 2S_{-1} = 0$. Those who used a similar method for S_{-2} were usually successful. Those who attempted to remember a formula often missed a term or used a wrong sign.

NO. 10

A cubic equation $x^3 + bx^2 + cx + d = 0$ has real roots α, β and γ such that

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{5}{12},$$

$$\alpha\beta\gamma = -12,$$

$$\alpha^3 + \beta^3 + \gamma^3 = 90.$$

i. Find the values of c and d. [3]

ii. Express $\alpha^2 + \beta^2 + \gamma^2$ in terms of b. [2]

iii. Show that $b^3 - 15b + 126 = 0$. [4]

iv. Given that $3 + i\sqrt{(12)}$ is a root of $y^3 - 15y + 126 = 0$, deduce the value of b. [2]

A level FURTHER MATHEMATICS 2019/Summer/Paper1_3/Q9

Mark scheme

Question	Answer	Marks	Guidance
i	$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\alpha\gamma}{\alpha\beta\gamma}\Rightarrow c = \alpha\beta + \beta\gamma + \alpha\gamma = 5$	M1 A1	Uses $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\alpha\gamma}{\alpha\beta\gamma}$.
	d =12	B1	
		3	
ii	$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$	M1	
	$(\alpha + \beta + \gamma)^2 - 10 = b^2 - 10$	A1	
	Alternative method for 9(ii)		
	$90 + bS_2 - 5b + 36 = 0 \Rightarrow S_2 = \frac{5b-126}{b}$	M1 A1	Sums over roots
		2	
iii	$x^3 + bx^2 + 5x + 12 = 0$ $(-b = \alpha + \beta + \gamma)$	M1	Formulates equation.
	$90 + b(b^2 - 10) - 5b + 36 = 0$	M1 A1	Sums over roots and uses **9(ii)**
	$b^3 - 15b + 126 = 0$	A1	
	Alternative method 1 for 9(iii)		
	$(\alpha + \beta + \gamma)^3 = \alpha^3 + \beta^3 + \gamma^3 + 3(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \alpha\gamma) - 3\alpha\beta\gamma$	M1 A1	Uses expansion of $(\alpha + \beta + \gamma)^3$
	$-b^3 = 90 - 15b + 36$	(M1)	Substitutes known values.
	$b^3 - 15b + 126 = 0$	A1	
	Alternative method 2 for 9(iii)		
	$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) = b^2 - 10$	M1 A1	Finds sum of squares in terms of b.

	$\frac{5b-126}{b} = b^2 - 10 \Rightarrow b^3 - 15b + 126 = 0$	M1 A1	Equates expressions.
		4	
iv	Real root is $b = -6$	M1 A1	Gives the real root for b.

Explanation

i)

$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\beta r+\alpha r+\alpha\beta}{\alpha\beta\gamma}$

From equation $x^3 + bx^2 + cx + d = 0$

We know that:$\alpha\beta\gamma = -d = -12, \alpha\beta + \alpha\gamma + \beta\gamma = c$ **M1**

d=12 **B1**

$\frac{\beta r+\alpha r+\alpha\beta}{\alpha\beta\gamma}=\frac{c}{-12}=-\frac{5}{12}$

C=5 **A1**

ii)

From equation $x^3 + bx^2 + 5x + 12 = 0$

$\alpha + \beta + \gamma = -b$

Then use formula:

$(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ **M1**

$(-b)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2 \times 5$

$\alpha^2 + \beta^2 + \gamma^2 = b^2 - 10$ **A1**

iii)

are roots of equation $x^3 + bx^2 + 5x + 12 = 0$ **M1**

$$\alpha^3 + b\alpha^2 + 5\alpha + 12 = 0$$

$$\sum\alpha^3 + b\Sigma\alpha^2 + 5\Sigma\alpha + 12\sum 1 = 0$$

Substitute known values:

$$90 + b(b^2 - 10) + 5(-b) + 36 = 0$$ **M1 A1**

$$b^3 - 15b + 126 = 0$$ **A1**

iv)

$y^3 - 15y + 126 = 0$ is a real coefficient equation

if $3 + i\sqrt{12}$ is a root, then $3 - i\sqrt{12}$ is also a root

the third root is b.

$$\left(3 + i\sqrt{12}\right) + \left(3 - i\sqrt{12}\right) + b = 0$$ (Coefficient of y^2)

b=-6 **M1 A1**

Comment

(i) (ii) These parts were very well done by most candidates. They recalled the relationships between the roots and coefficients of the cubic equation and were able to use them accurately.

(iii) Those who used the cubic equation given in the question were usually successful, though a small number forgot to find S_0 . Errors were seen when candidates did not recall the formula for $\sum\alpha^3$ correctly.

(iv) The most common approach was to work with the product of roots, realising that complex roots occur in conjugate pairs. Some candidates did not specify that b is the real root.